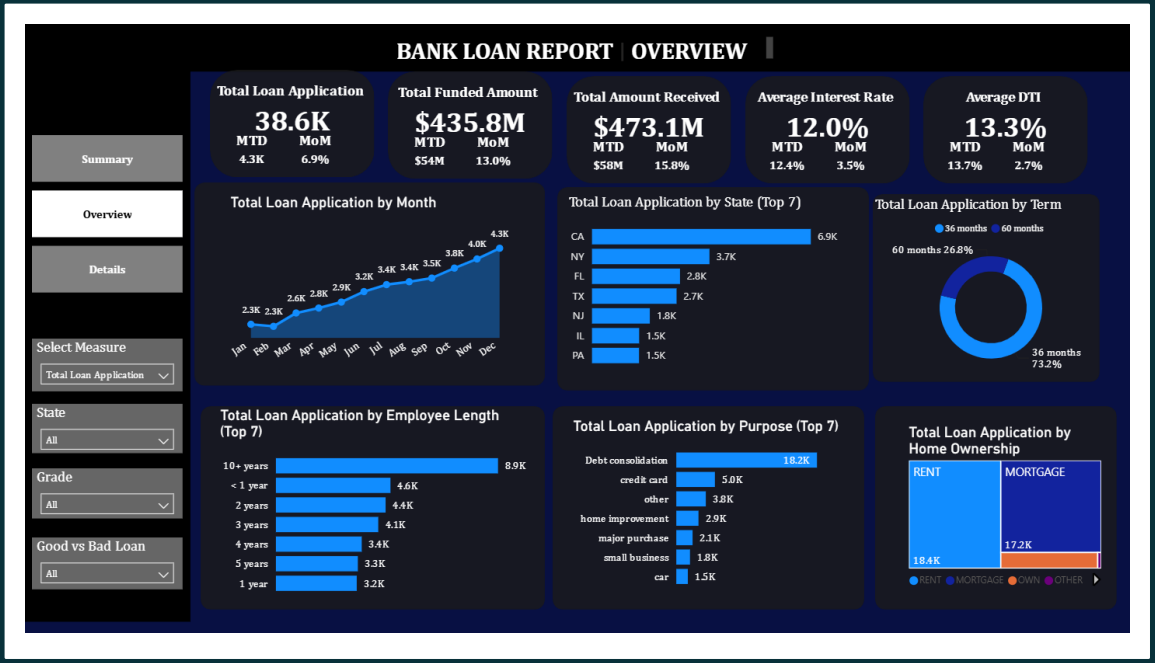
POWER BI CASE STUDY

Bank Loan Performance & Risk Analysis

KPIs, good vs bad loans and repayment



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Tools: Power BI, DAX | Status: For review

Independent portfolio project.

INDEX

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Thirteen pages from the headline numbers to the recommendations.

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FINDINGS

Executive summary

Four points a manager needs from this report. Average loan size and recovery rates are calculated from the Summary table. Amounts in USD.

$37.3M

$435.8M funded, $473.1M received

38.6K loan applications in 2021 returned a net gain, 8.6% above the funded amount.

15.0%

Bad loans: 13.8% of loans, 15.0% of money

5.3K Charged Off loans, $65.5M funded. Their average loan ($12.3K) is above Fully Paid ($10.9K).

56.9%

Charged Off loans paid back only 56.9%

$37.3M received on $65.5M funded, a $28.2M gap. Fully Paid loans returned 17.1% above funded.

+87%

Volume almost doubled; risk rose in December

Applications grew from 2.3K to 4.3K. December rate 12.4% and DTI 13.7% sit above the full-year 12.0% and 13.3%.

SCOPE

Scope and data

Method: define KPIs with MTD and MoM, split good and bad loans, then break the book down by state, purpose, term, tenure and home ownership. Limit: one year of data, no income or credit score.

38,576

loan applications issued in 2021

$435.8M

total funded amount (USD)

3 pages

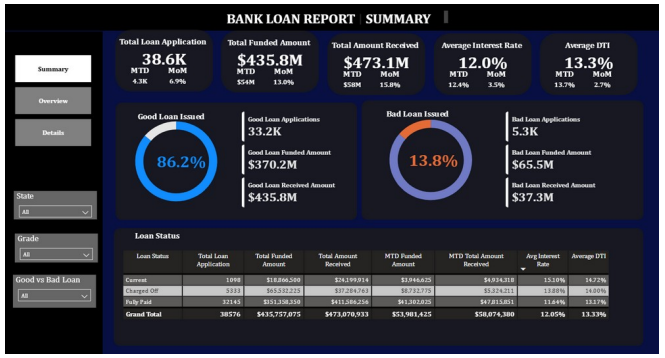
Summary, Overview and Details

1. How is the loan book performing: volume, funding, repayment, rate and DTI?
2. What share of loans are good or bad, and how much money does each bring back?
3. Where do applications come from: state, purpose, term, tenure, home ownership?
4. How did volume change month by month, and what does the latest month signal?

DASHBOARD

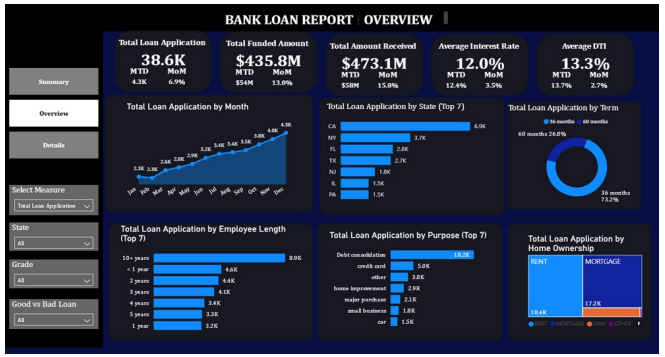
The Power BI dashboard

Slicers: State, Grade and Good vs Bad Loan on every page, plus a Select Measure switch on Overview and Details.



Summary

KPI cards, good vs bad loan split and the loan status table.



Overview

Monthly trend plus top 7 by state, term, tenure, purpose and home ownership.



Details

Loan-level table: grade, sub grade, rate, installment and amount received.

KPIS

Five KPIs, each with MTD and MoM

Month-to-date (December) and the change against November.

Loan applications	38.6K	MTD 4.3K	MoM +6.9%
Funded amount	$435.8M	MTD $54M	MoM +13.0%
Amount received	$473.1M	MTD $58M	MoM +15.8%
Average interest rate	12.0%	MTD 12.4%	MoM +3.5%
Average DTI	13.3%	MTD 13.7%	MoM +2.7%

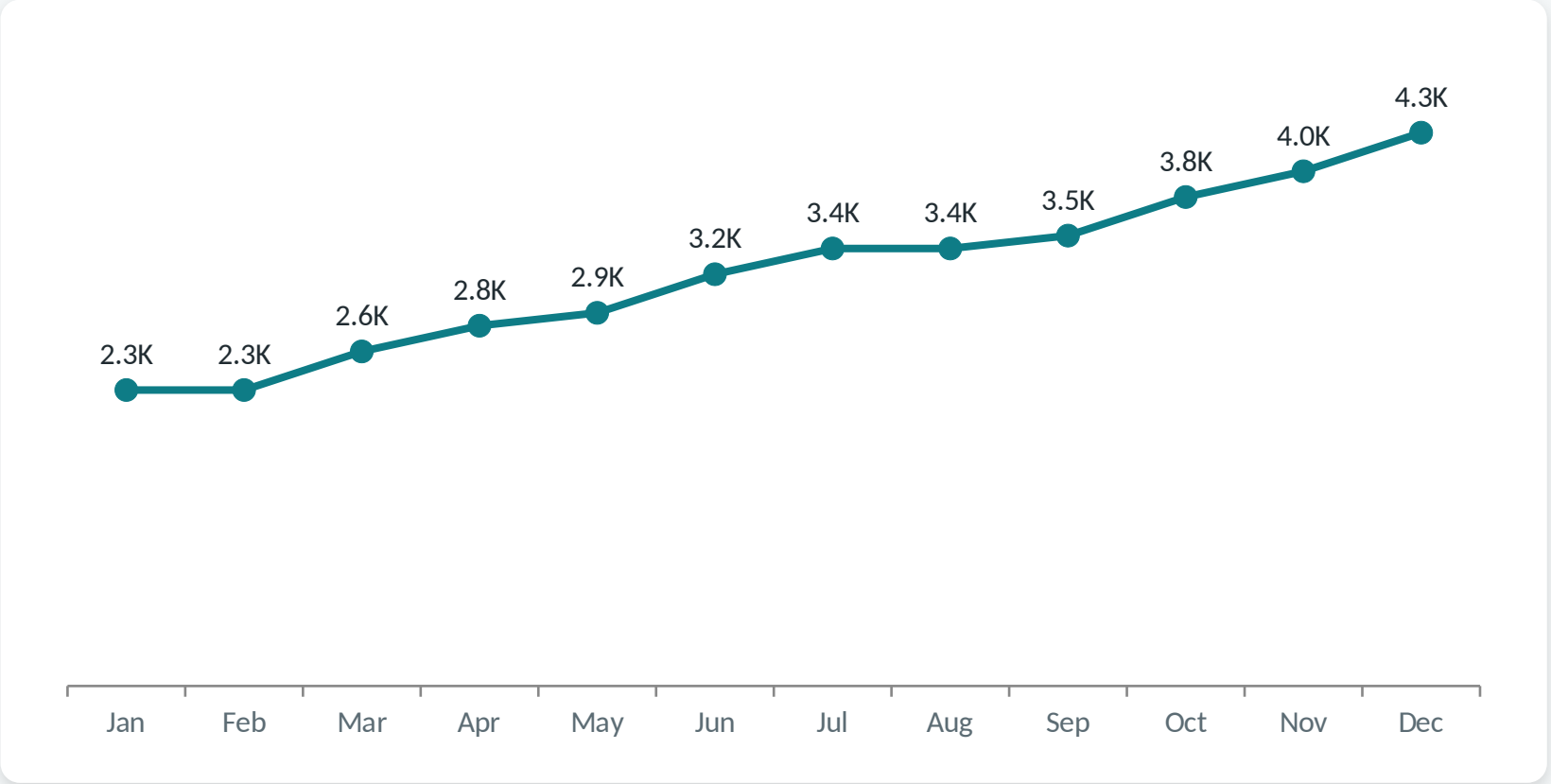
What this means

- Received exceeds funded by $37.3M (8.6%), so the book is profitable overall
- December funded amount grew 13.0% MoM, faster than applications (6.9%), so loans are getting bigger
- Average funded amount is about $11.3K per loan, about $12.5K in December (calculated)

TREND

Applications almost doubled in 2021

Total loan applications by month, in thousands. December MoM growth is 6.9%.



2.3K to 4.3K

January to December

+87%

growth over the year

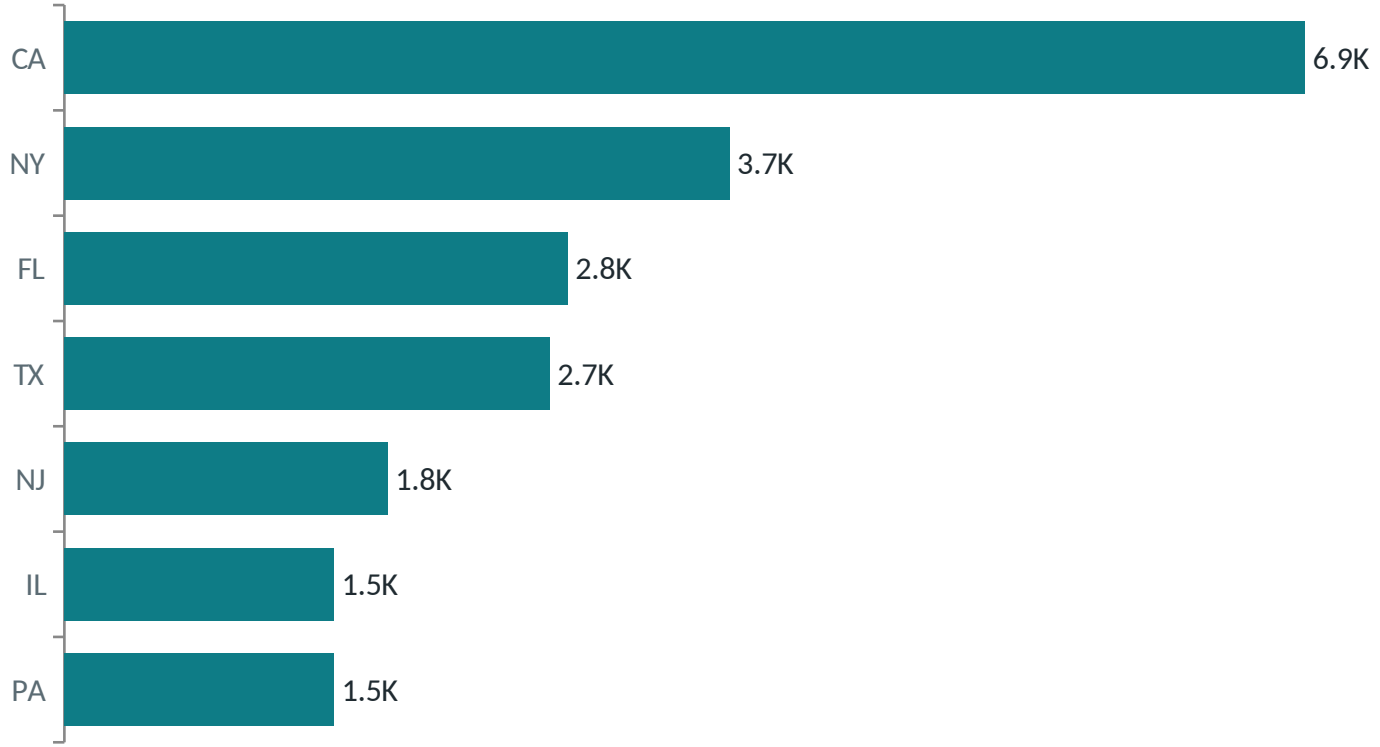
9 + 2

9 rises and 2 flat months (Feb, Aug); no month fell

STATES

California alone is 17.9% of applications

Total loan applications by state, top 7, in thousands. The top 7 states hold 54% of all applications.



17.9%

of applications are from CA (6.9K)

42%

from the top 4: CA, NY, FL, TX

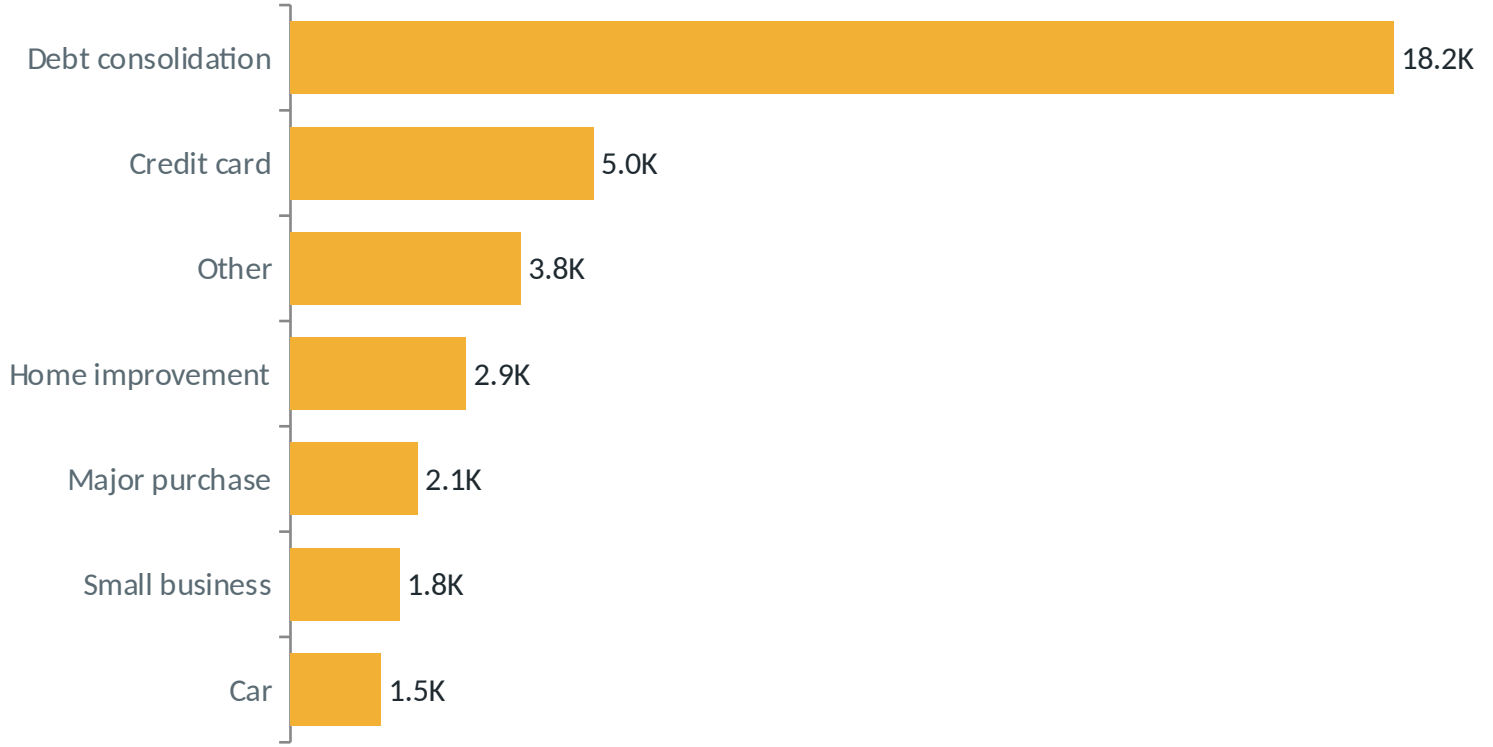
1.9x

CA against NY, the second state

PURPOSE

Nearly half of loans are for debt consolidation

Total loan applications by purpose, top 7, in thousands.



47%

of applications: debt consolidation (18.2K)

60%

with credit card refinancing (23.2K)

1.5K

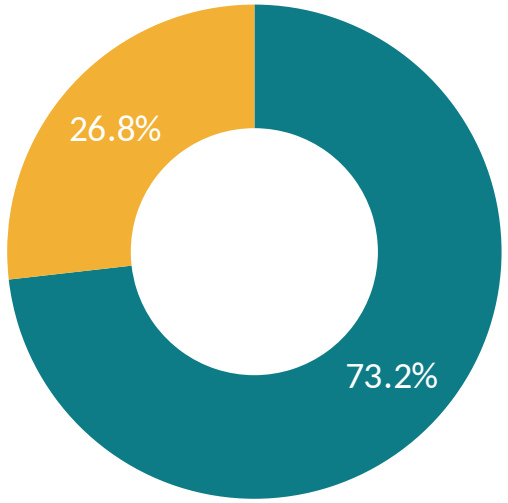
car loans, the smallest of the top 7

PROFILE

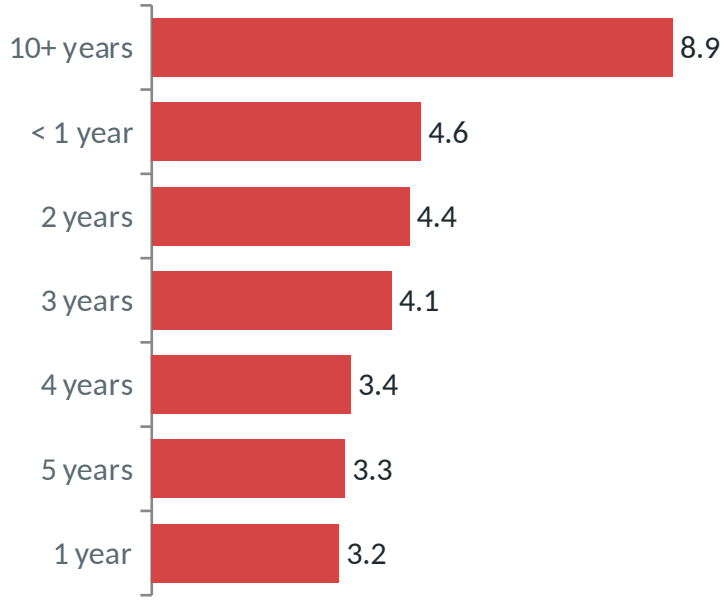
Borrower profile: term, tenure and home

10+ years of employment is 23% of applications. OWN and OTHER together are about 3.0K, or 7.7% (calculated).

Loan term



Employment length (K)



18.4K

RENT: 47.7% of applications

17.2K

MORTGAGE: 44.6% of applications

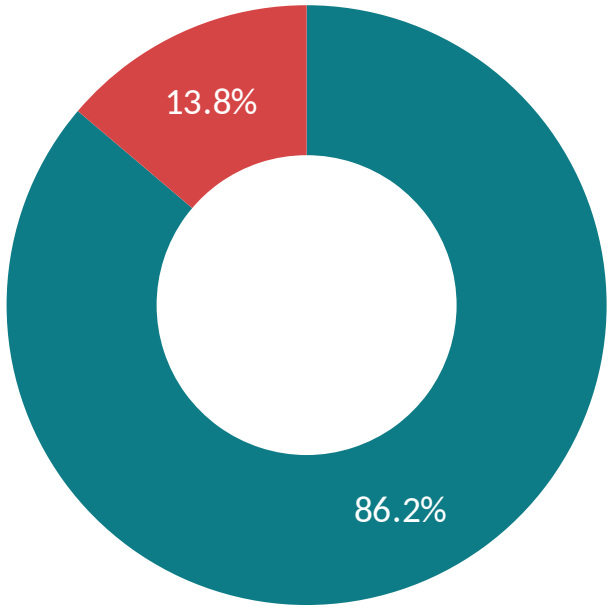
~3.0K

OWN and OTHER: 7.7%

RISK

Bad loans are 13.8% of loans but 15.0% of money

Good loans are Fully Paid and Current. Bad loans are Charged Off.



Metric	Good loans	Bad loans
Applications	33.2K	5.3K
Funded	$370.2M	$65.5M
Received	$435.8M	$37.3M

Why it matters

- Bad loans are 15.0% of funded money ($65.5M of $435.8M), above their 13.8% share of loans
- Their average loan is about $12.3K, against $10.9K for Fully Paid loans

STATUS

Charged Off loans paid back only 56.9%

Current loans are still paying, so their result is not final.

Status	Loans	Funded	Received	Vs funded	Rate	DTI
Fully Paid	32,145	$351.4M	$411.6M	+17.1%	11.64%	13.17%
Current	1,098	$18.9M	$24.2M	+28.3%	15.10%	14.72%
Charged Off	5,333	$65.5M	$37.3M	-43.1%	13.88%	14.00%
Total	**38,576**	**$435.8M**	**$473.1M**	+8.6%	**12.05%**	**13.33%**

$28.2M

gap on Charged Off loans

+2.2 pts

Charged Off rate over Fully Paid (13.88% vs 11.64%)

+0.8 pts

Charged Off DTI over Fully Paid (14.00% vs 13.17%)

DETAILS

Details: the biggest returns are high-rate loans

Caution: these loans return most when repaid. The dashboard does not yet show their Charged Off rate.



ID	Purpose	Home Ownership	Grade	Sub Grade	Issue Date	Funded Amount	Int Rate	Installment	Amount Received
989285	Debt consolidation	RENT	G	G1	Monday, October 11, 2021	$35,000	22.7%	$981.5	$58,564
812976	Debt consolidation	MORTGAGE	G	G2	Wednesday, August 11, 2021	$35,000	22.5%	$976.2	$58,480
972576	credit card	MORTGAGE	F	F5	Monday, October 11, 2021	$35,000	22.3%	$973.6	$57,835
874599	Debt consolidation	MORTGAGE	G	G3	Saturday, September 11, 2021	$35,000	22.8%	$983.7	$56,849
768930	small business	MORTGAGE	F	F3	Friday, June 11, 2021	$35,000	21.0%	$946.7	$56,663
674448	Debt consolidation	MORTGAGE	G	G2	Thursday, February 11, 2021	$35,000	20.5%	$936.7	$56,199
914211	Debt consolidation	MORTGAGE	F	F1	Monday, October 11, 2021	$35,000	20.9%	$944.7	$55,907
772157	small business	RENT	G	G1	Friday, June 11, 2021	$35,000	22.1%	$968.9	$55,769
1057770	Debt consolidation	MORTGAGE	E	E5	Saturday, December 11, 2021	$35,000	20.3%	$933.1	$55,139
833224	Debt consolidation	MORTGAGE	F	F2	Wednesday, August 11, 2021	$35,000	20.6%	$939.4	$55,106
698163	home improvement	MORTGAGE	G	G2	Thursday, March 11, 2021	$35,000	20.5%	$936.7	$54,774
762870	Debt consolidation	OWN	E	E4	Tuesday, May 11, 2021	$35,000	19.3%	$913.5	$54,746
768153	home improvement	MORTGAGE	F	F2	Friday, June 11, 2021	$35,000	20.6%	$939.4	$54,715
1057239	Debt consolidation	RENT	E	E3	Saturday, December 11, 2021	$35,000	19.4%	$916.0	$54,427
1034299	credit card	MORTGAGE	E	E3	Saturday, December 11, 2021	$35,000	19.4%	$916.0	$54,287
1008529	Debt consolidation	MORTGAGE	E	E5	Thursday, November 11, 2021	$35,000	20.3%	$933.1	$54,182
733401	credit card	RENT	G	G4	Sunday, April 11, 2021	$35,000	21.2%	$951.2	$54,132
821705	Debt consolidation	MORTGAGE	E	E4	Sunday, July 11, 2021	$35,000	19.3%	$913.5	$54,112
970684	Debt consolidation	RENT	F	F4	Monday, October 11, 2021	$35,000	19.8%	$925.5	$54,005

- Top rows: $35,000 loans, grades E to G, rates about 19% to 23%
- Best row: ID 989285, G1 at 22.7%, $58,564 received on $35,000 (+67%)
- Mostly debt consolidation, with mortgage or rent

LIMITS

Limits of the data

Four limits to keep in mind when reading the results.

1

One year of data

All loans are from 2021, so there is no multi-year trend.

2

Current loans are not final

1,098 loans are still paying, so their result may change.

3

No income or credit score

The dashboard shows differences, not why borrowers default.

4

Some figures are calculated

Average loan size and recovery rates come from the Summary table.

NEXT STEPS

Recommendations

Five actions, in the order I would take them.

1. **Review credit rules for higher-rate, higher-DTI loans**
   Charged Off loans sit higher on both rate and DTI.

2. **Track bad loans in dollars, not only count**
   They are 15.0% of money against 13.8% of loans.

3. **Watch December risk signals**
   Rate 12.4% and DTI 13.7% rose above the full year while volume grew.

4. **Set risk checks for debt consolidation and credit card**
   Together they are 60% of applications.

5. **Add default rate by grade to the dashboard**
   Check E to G loans before treating high returns as safe.

THANK YOU

Questions?



Happy to walk through the Power BI report, or build the same analysis on your own data.

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Independent portfolio project. Figures come from the Power BI dashboard (38,576 loans, 2021, USD).